

# When Does a Random-Scale Flight Path Look Lognormal?

A spherical bridge note from OpenSky trajectories

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## Abstract

An hour of globally sampled aircraft positions produces a tempting picture: RMS departures from endpoint great-circle arcs are positive, strongly right-skewed, and superficially lognormal. We turn that visual hypothesis into a geometric model. Each deviation is factored as  $D = AQ$ , where  $A$  sets route scale and  $Q$  is the RMS shape of an endpoint-pinned bridge. We prove that an independent lognormal scale suppresses the non-Gaussian cumulants of  $\log Q$  at explicit rates and give a Wasserstein convergence criterion. For a Brownian bridge, the criterion yields a concrete threshold. Yet the observed log-skewness and kurtosis sharply contradict the resulting single-population prediction. The same theorem that explains a lognormal-looking center therefore diagnoses a distinct near-geodesic boundary population, and a sequential small-ball test gives a retrospective, in-sample estimate of when that assignment becomes visible. The result is not a universal lognormal law, but a route from visual regularity to a falsifiable mixture model and a time-to-diagnosis statistic.

## A visual regularity with a geometric question

The OpenSky Network makes large-scale ADS-B trajectory research possible [1]. We sampled 60 one-minute global state snapshots, retained aircraft with all 60 positions, required at least 25 km between endpoints, and removed any track with a one-minute jump above 30 km. Of 12,524 observed identifiers, 2,183 had valid positions in every snapshot; the final filtered cohort contains 1,780 tracks. Continuous visibility is receiver- and geography-dependent and favors aircraft that remain observable for the full hour. Thus every population fraction below is conditional on this coverage-selected cohort, not an estimate for global aviation.

For aircraft  $i$ , write  $x_i(t) \in S_R^2$  for its position on the spherical Earth at normalized time  $t \in [0, 1]$ . The direct reference is the minor endpoint geodesic, a standard spherical construction in directional statistics [2],

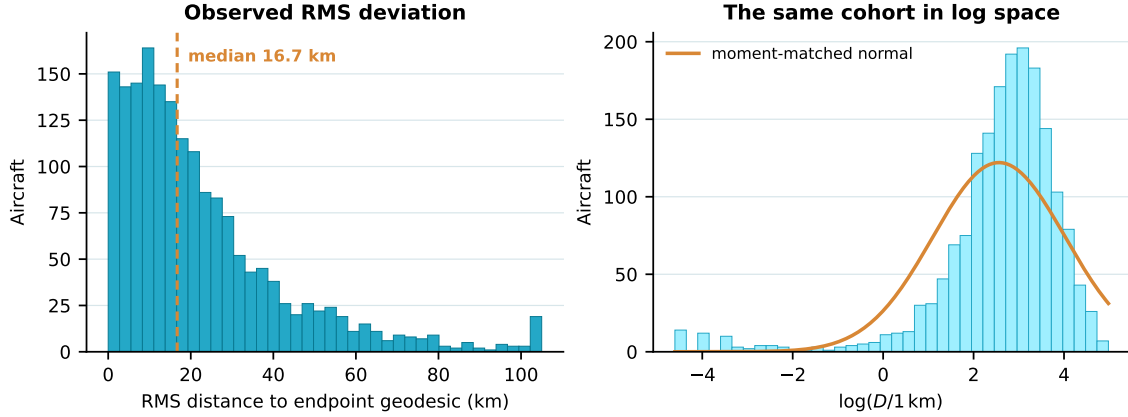
$$g_i(s) = \text{Exp}_{x_i(0)}\left(s \text{Log}_{x_i(0)} x_i(1)\right), \quad s \in [0, 1]. \quad (1)$$

Here  $s$  is dimensionless progress, and  $g_i(s)$  is a position rather than a distance. If  $d_{S_R^2}$  is spherical distance in kilometres, define cross-track displacement and its sampled RMS by

$$\eta_i(t) = d_{S_R^2}(x_i(t), g_i([0, 1])), \quad D_i = \left[ \frac{1}{60} \sum_{j=1}^{60} \eta_i(t_j)^2 \right]^{1/2}. \quad (2)$$

The median  $D_i$  is 16.7 km and the mean is 22.7 km. Figure 1 shows why a lognormal model is initially plausible and why that impression is incomplete.

Classical lognormal modeling begins with  $\log(D/d_0) \sim N(\mu, \sigma^2)$  for reference distance  $d_0 = 1$  km [3]. That describes a shape but supplies no route geometry. We instead ask: *what mechanism makes a spherical path statistic look lognormal, what would falsify it, and how long must one watch before the falsifying boundary component is recognizable?*



**Figure 1:** RMS spherical deviation for the quality-filtered cohort. Left: the application histogram, truncated visually at the 99th percentile while retaining tail observations in the end bin. Right: the 1,776 positive values in log space with a moment-matched Gaussian. The broad center looks compatible with lognormality, but the long lower tail does not.

#### Narrative arc.

- **A direct reference.** Aircraft rarely follow the shortest spherical path between observed endpoints, so the endpoint geodesic is a reference rather than an operational route.
- **Scale and shape.** Each RMS deviation is written as  $D = AQ$ :  $A$  sets overall magnitude, while  $Q$  captures standardized endpoint-pinned shape.
- **The smoothing prediction.** Broad variation in  $A$  should suppress the skewness and kurtosis contributed by  $Q$ , making the Brownian product nearly Gaussian in log space.
- **The boundary population.** The observed log-skewness is  $-2.40$ , with 53 tracks at or below 0.1 km RMS deviation forming a distinct lower-tail boundary.
- **Sequential diagnosis.** After the mixture is forced, prefix residuals and Wasserstein gaps give an in-sample, retrospective estimate of how long one must watch before assigning a single aircraft to that boundary component.

### Scale, shape, and the bridge factor

Factor the observed departure into a kilometre-valued scale and a dimensionless shape,

$$\eta_i(t) = A_i Z_i(t), \quad D_i = A_i Q_i, \quad Q_i = \left[ \int_0^1 Z_i(t)^2 dt \right]^{1/2}, \quad (3)$$

where  $Z_i(0) = Z_i(1) = 0$ . Intuitively,  $A_i$  asks *how large* the route departure is, while  $Z_i$  asks *what shape* it takes between fixed endpoints. Across a population, independent densities  $f_A$  and  $f_Q$  induce the product law

$$f_D(d) = \int_0^\infty f_A(d/q) f_Q(q) \frac{dq}{q}. \quad (4)$$

The Jacobian  $1/q$  is the signature of multiplicative rather than additive mixing. A single lognormal is therefore not the starting mechanism; it is a possible approximation to this geometry-driven product.

### When the product becomes lognormal

Set  $U_\ell = \log(A_\ell/d_0)$  and  $V_\ell = \log Q_\ell$  for a sequence of scale-shape populations. The index  $\ell$  labels increasing scale dominance, not time or aircraft.

**Theorem 1** (Lognormal smoothing). *Suppose  $U_\ell \sim N(\mu_\ell, \sigma_\ell^2)$  is independent of  $V_\ell$ , with  $m_\ell = \mathbb{E}V_\ell$  and  $\tau_\ell^2 = \text{Var}(V_\ell) < \infty$ . Define*

$$W_\ell = \frac{\log(D_\ell/d_0) - (\mu_\ell + m_\ell)}{\sqrt{\sigma_\ell^2 + \tau_\ell^2}}.$$

If  $\tau_\ell^2/(\sigma_\ell^2 + \tau_\ell^2) \rightarrow 0$ , then  $W_\ell \Rightarrow N(0, 1)$ . More precisely, for  $G_\ell \sim N(m_\ell, \tau_\ell^2)$ ,

$$d_{W_2}(W_\ell, N(0, 1)) \leq \frac{d_{W_2}(V_\ell, G_\ell)}{\sqrt{\sigma_\ell^2 + \tau_\ell^2}},$$

where  $d_{W_2}$  is quadratic Wasserstein distance [4].

*Proof.* Replace  $V_\ell$  by its Gaussian moment match  $G_\ell$ . Then  $U_\ell + G_\ell$  is exactly Gaussian with the displayed centering and variance. Convolution with the same independent  $U_\ell$  contracts  $W_2$ . Finally, independent coupling gives  $d_{W_2}(V_\ell, G_\ell) \leq \sqrt{2}\tau_\ell$ , so the stated variance ratio forces convergence.  $\square$

The theorem formalizes a simple picture: if fleet-to-fleet variation in route scale is broad enough, it blurs the finer distributional signature of route shape.

**Proposition 1** (Exact cumulant attenuation). *For one fixed population, let  $\rho = \tau_Q^2/(\sigma_A^2 + \tau_Q^2)$ , where  $\sigma_A^2 = \text{Var}(\log(A/d_0))$  and  $\tau_Q^2 = \text{Var}(\log Q)$ . For every  $r \geq 3$ ,*

$$\kappa_r(W) = \frac{\kappa_r(\log Q)}{(\sigma_A^2 + \tau_Q^2)^{r/2}}.$$

Consequently,

$$\text{Skew}(W) = \text{Skew}(\log Q)\rho^{3/2}, \quad \text{ExKurt}(W) = \text{ExKurt}(\log Q)\rho^2.$$

*Proof.* Independent cumulants add, and a Gaussian has no cumulants above order two. Standardizing by the total log-variance gives the first identity; writing the third and fourth cumulants in standardized form gives the second line.  $\square$

These identities turn “approximately lognormal” into a checkable statement: shape-induced skewness decays as  $\rho^{3/2}$  and excess kurtosis as  $\rho^2$ .

## The Brownian benchmark and its failure

A standard Brownian bridge is the simplest endpoint-pinned random shape. Its Karhunen–Loève expansion and RMS functional are

$$Z(t) = \sum_{k=1}^{\infty} \frac{\sqrt{2}\xi_k}{\pi k} \sin(\pi kt), \quad Q^2 = \int_0^1 Z(t)^2 dt = \sum_{k=1}^{\infty} \frac{\xi_k^2}{\pi^2 k^2}, \quad (5)$$

with independent  $\xi_k \sim N(0, 1)$ . This is the classical quadratic Brownian-bridge functional underlying Cramér–von Mises theory [5]; its Laplace transform is

$$\mathbb{E}e^{-\lambda Q^2} = \left( \frac{\sqrt{2\lambda}}{\sinh \sqrt{2\lambda}} \right)^{1/2}, \quad \lambda > 0. \quad (6)$$

A two-million-draw, 500-mode numerical evaluation gives

$$\text{SD}(\log Q) \approx 0.385, \quad \text{Skew}(\log Q) \approx 0.178, \quad \text{ExKurt}(\log Q) \approx -0.348.$$

Substitution in the proposition yields a concrete conclusion:  $\sigma_A \gtrsim 0.50$  pushes both surviving standardized cumulants below 0.05 in magnitude. A moderately broad lognormal scale should therefore make this Brownian product look very nearly lognormal.

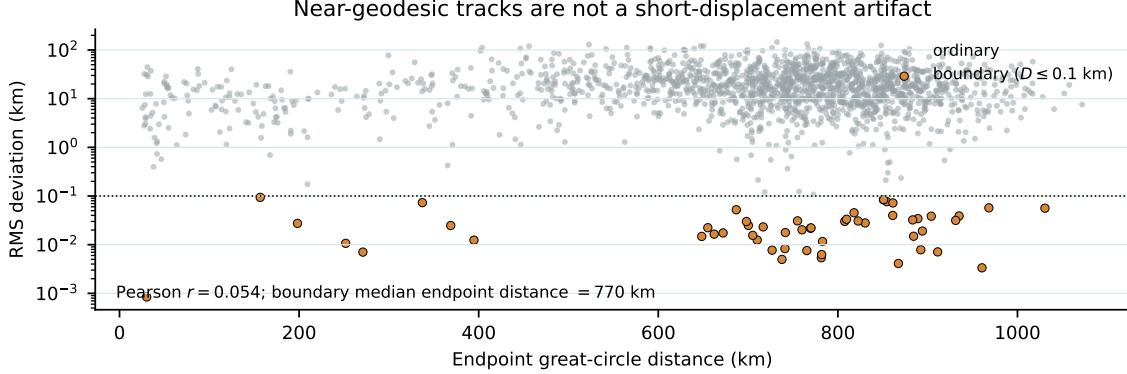
The observed cohort supplies a sharp diagnostic. Among the 1,776 positive  $D_i$  values,

$$\text{SD}(\log(D/d_0)) = 1.465, \quad \text{Skew} = -2.40, \quad \text{ExKurt} = 8.11.$$

Variance matching under the independent Brownian model implies  $\sigma_A \approx 1.413$ . At that scale spread, the proposition predicts skewness about 0.003 and excess kurtosis about  $-0.002$ —essentially Gaussian in log space. The data are nowhere close. The mismatch is driven by a near-geodesic boundary: 53 tracks have recorded RMS deviation at or below 0.1 km.

### Route-length and filter diagnostics

The boundary is not produced by admitting short tracks. Figure 2 plots RMS deviation against endpoint distance: their Pearson correlation is only 0.054, the boundary median is 770 km versus 752 km for ordinary tracks, and only one of the 53 boundary tracks has an endpoint distance below 100 km.



**Figure 2:** Endpoint-distance diagnostic for the 1,780-track cohort. Boundary tracks span the same route-length range as the ordinary population, ruling out short endpoint displacement as the source of the lower-tail boundary.

Holding complete coverage and the 30 km jump screen fixed, changing the minimum endpoint distance leaves the headline statistics nearly unchanged:

| Cut (km) | <i>n</i> | $D \leq 0.1$ | Median | Mean  | P90   | P99    | Skew  | Ex. kurt. |
|----------|----------|--------------|--------|-------|-------|--------|-------|-----------|
| 10       | 1801     | 53           | 16.62  | 22.61 | 50.04 | 102.30 | -2.38 | 8.05      |
| 25       | 1780     | 53           | 16.70  | 22.66 | 50.06 | 102.61 | -2.40 | 8.11      |
| 100      | 1729     | 52           | 16.99  | 22.95 | 50.84 | 103.37 | -2.45 | 8.33      |

Here  $D$  and its percentiles are in kilometres; skewness and excess kurtosis refer to positive values of  $\log(D/d_0)$ . This sensitivity check supports the boundary finding, but it does not remove the full-hour coverage conditioning.

### From an attractive fit to a useful rejection

The mathematical arc reverses the usual modeling order. We did not choose a flexible mixture merely because a histogram fit was imperfect. Geometry first proposed  $D = AQ$ ; the smoothing theorem then showed that the Brownian single-regime model should be *more* Gaussian in log space than the data; only then did a mixture become necessary.

A natural next model is the finite mixture form [6]

$$f_D(d) = \pi_0 f_0(d) + \sum_{c=1}^C \pi_c f_D(d | c), \quad \pi_c \geq 0, \quad \sum_{c=0}^C \pi_c = 1, \quad (7)$$

where  $f_0$  represents the near-geodesic boundary and the continuous components represent distinct route-shape or flight-phase regimes. The current hour cannot identify all such classes, and endpoint geodesics are geometric counterfactuals rather than operationally feasible routes. Those limitations are substantive. Still, the note establishes a clean diagnostic principle: broad lognormal scaling can hide ordinary bridge-shape nonnormality, so a large residual departure is evidence for heterogeneity, dependence, or a boundary mechanism—not merely a slightly misspecified bridge.

### Sequential diagnosis of the boundary population

Once the boundary component has been detected at the cohort level, the next question is individual and temporal: how long must one watch a trajectory before assigning it to that component? Let  $C = 0$  denote the near-geodesic boundary population and  $C = 1$  the ordinary bridge population. For the first  $n$  samples define

the prefix residual

$$R_n^2 = \frac{1}{n} \sum_{j=1}^n \eta(t_j)^2, \quad \mathcal{F}_n = \sigma\{\eta(t_1), \dots, \eta(t_n)\}. \quad (8)$$

For a fixed tolerance  $\tau$ , persistent small residuals are informative only when they are rare under the ordinary component.

**Theorem 2** (Sequential small-ball diagnosis). *Let  $\pi_0 = \mathbb{P}(C = 0)$ . Suppose that, for some  $\tau > 0$ ,*

$$\mathbb{P}_1(R_n \leq \tau) \leq e^{-nI(\tau)} \quad \text{and} \quad \mathbb{P}_0(R_n \leq \tau) \geq 1 - \beta_n.$$

Then

$$\mathbb{P}(C = 0 \mid R_n \leq \tau) \geq \frac{\pi_0(1 - \beta_n)}{\pi_0(1 - \beta_n) + (1 - \pi_0)e^{-nI(\tau)}}.$$

In particular, posterior confidence  $1 - \alpha$  is reached once

$$n \gtrsim \frac{\log((1 - \pi_0)/\pi_0) + \log((1 - \alpha)/\alpha)}{I(\tau)}$$

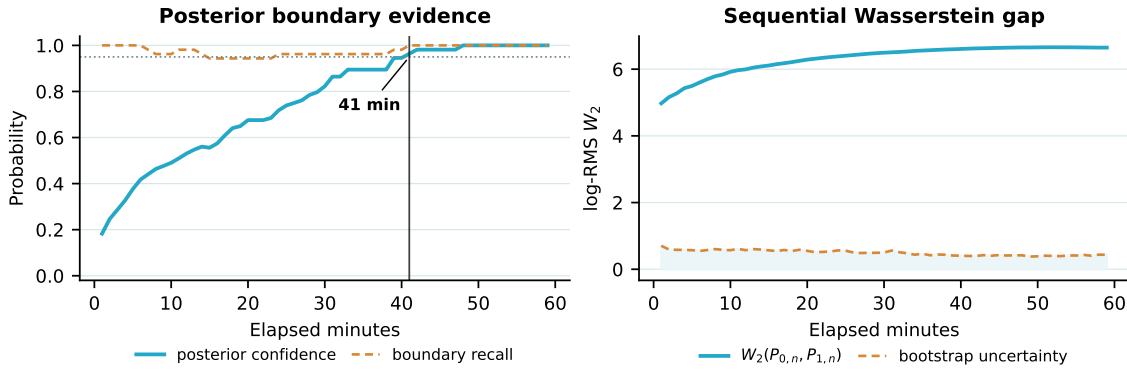
up to the boundary miss term  $\beta_n$ .

*Proof.* Bayes' rule gives

$$\mathbb{P}(C = 0 \mid R_n \leq \tau) = \frac{\pi_0 \mathbb{P}_0(R_n \leq \tau)}{\pi_0 \mathbb{P}_0(R_n \leq \tau) + (1 - \pi_0) \mathbb{P}_1(R_n \leq \tau)}.$$

The two displayed assumptions give the bound. Solving the same inequality for  $n$  gives the stated waiting-time scale.  $\square$

This is the time-series version of the mixture conclusion: not every straight-looking prefix is diagnostic, but persistent near-zero residual energy becomes decisive once ordinary bridges have too little small-ball mass left.



**Figure 3:** Retrospective, in-sample boundary diagnostics using the final endpoint and full-hour boundary labels. Left: empirical posterior confidence and boundary recall for  $R_n \leq 0.1$  km. The prior boundary mass is 53/1780, so confidence becomes high only after persistence. Right: the empirical Wasserstein gap between the log-prefix RMS laws, plotted against a bootstrap uncertainty floor.

The same idea can be expressed in the transport language already used in the smoothing theorem. Let  $Y_n = \log(R_n/d_0)$ , and let  $P_{0,n}$  and  $P_{1,n}$  be the laws of  $Y_n$  under the boundary and ordinary components. Define

$$\Delta_n = W_2(P_{0,n}, P_{1,n}).$$

A conservative critical time is

$$T_W = \inf\{n : \Delta_n > r_{0,n}(\alpha) + r_{1,n}(\alpha) + m\},$$

where  $r_{c,n}(\alpha)$  are sampling radii and  $m$  is a chosen margin. For one observed aircraft with value  $y_n$ , the point-to-law score

$$S_n(y_n) = W_2^2(\delta_{y_n}, P_{1,n}) - W_2^2(\delta_{y_n}, P_{0,n})$$

is positive when the observation is closer to the boundary law than to the ordinary law. Thus the practical stopping rule combines individual posterior evidence with population separation:

$$T_* = \max\{T_{\text{post}}, T_W\}, \quad T_{\text{post}} = \inf\{n : \mathbb{P}(C = 0 \mid \mathcal{F}_n) \geq 1 - \alpha\}.$$

In the present hour, with  $\tau = 0.1$  km and  $\pi_0 = 53/1780$ , the final-endpoint diagnostic crosses 0.95 posterior confidence after roughly forty-one elapsed minutes. This threshold is retrospective and in-sample: both the reference geodesic and the boundary labels use the completed 60-minute track, and no later or held-out horizon validates the assignment. It should therefore be read as a descriptive critical time for this capture, not a prospective performance claim. A genuinely online rule must estimate the destination-side reference without the final endpoint and be tested on an independent, longer window; many ordinary trajectories are locally straight, so that problem is harder.

## References

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- [1] M. Schäfer, M. Strohmeier, V. Lenders, I. Martinovic, and M. Wilhelm, “Bringing up OpenSky: A large-scale ADS-B sensor network for research,” *Proc. 13th ACM/IEEE IPSN*, pp. 83–94, 2014. doi:10.1109/IPSIN.2014.6846743.
  - [2] K. V. Mardia and P. E. Jupp, *Directional Statistics*. Wiley, 2000. doi:10.1002/9780470316979.
  - [3] J. Aitchison and J. A. C. Brown, *The Lognormal Distribution*. Cambridge University Press, 1957.
  - [4] C. Villani, *Optimal Transport: Old and New*. Springer, 2009. doi:10.1007/978-3-540-71050-9.
  - [5] T. W. Anderson and D. A. Darling, “Asymptotic theory of certain goodness of fit criteria based on stochastic processes,” *Ann. Math. Statist.*, vol. 23, no. 2, pp. 193–212, 1952. doi:10.1214/aoms/1177729437.
  - [6] G. J. McLachlan and D. Peel, *Finite Mixture Models*. Wiley, 2000. doi:10.1002/0471721182.